

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		80% bioethanol : 20% petrol ☐ 20% bioethanol : 20% petrol ☐ 20% bioethanol : 80% petrol ☒			1	1		
		(ii)		the energy value decreases ☒ the density decreases ☐ the flash point decreases ☐			1	1		
		(iii)		award (1) for any sensible answer e.g. • fermentation releases carbon dioxide • building factories releases carbon dioxide • transporting bioethanol releases carbon dioxide • burning forests releases carbon dioxide • deforestation decreases the amount of carbon dioxide removed from the atmosphere link to carbon dioxide is essential			1	1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(b)	(i)		H H H-C-C-O-H H H		1		1		
		(ii)	I	magnesium ethanoate		1		1		
			II	Mg(CH ₃ COO) ₂	1			1		
	(c)			A B C copper(II) chloride iron(II) chloride ammonium sulfate iron(III) chloride copper(II) carbonate ammonium carbonate iron(II) sulfate			3	3		3
				Question 8 total	1	2	6	9	0	3